DSA0602 Data Handling and Visualization

Slot C

CO2 – Assessment Test 1 (AT1)

Data Interpretation Exercise

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1. X–Y Relationships — Scatter Plot

Question: Given data on years of work experience and monthly salary (₹) of 22 employees, construct an appropriate chart to explore this X–Y relationship.

Answer:

A **Scatter Plot** is the most appropriate visualization for representing the relationship between work experience and monthly salary.

- **X-axis:** Years of work experience.
- **Y-axis:** Monthly salary (₹).
- **Data points:** Each point represents one employee.

Explanation:

A scatter plot helps identify whether salary increases with work experience. If the points generally move upward from left to right, it indicates a positive relationship between experience and salary. It also helps identify unusual observations and employees whose salaries differ significantly from those of others with similar experience.

Conclusion: A scatter plot is suitable because it effectively displays the relationship between two numerical variables.

Scatter Plot of Work Experience and Monthly Salary

Create data for 22 employees

```
experience <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10,
1, 3, 5, 7, 9, 2, 4, 6, 8, 10, 5, 7)
```

```
salary <- c(15000, 18000, 22000, 25000, 28000, 32000,
35000, 38000, 42000, 45000, 16000, 23000,
30000, 36000, 43000, 19000, 26000, 33000,
40000, 47000, 29000, 37000)
```

Combine data into a data frame

```
employees <- data.frame(experience, salary)
```

Display employee data

```
print(employees)
```

Create scatter plot

```
plot(employees$experience, employees$salary,
main = "Relationship Between Work Experience and Salary",
xlab = "Years of Work Experience",
ylab = "Monthly Salary (Rs.)",
pch = 19,
col = "blue")
```

Add trend line

```
abline(lm(salary ~ experience, data = employees),
col = "red", lwd = 2)
```



2. Geospatial Data — Map-Based Chart

Question: Using data on the number of retail outlets of a company located in eight different cities, construct a map-based chart to visualize their geographic distribution.

Answer:

A **Proportional Symbol Map** is an appropriate visualization for displaying the geographic distribution of retail outlets across eight cities.

- **Geographic locations:** The eight cities.
- **Symbols:** Circles placed at each city's location.
- **Circle size:** Proportional to the number of retail outlets.

Explanation:

A proportional symbol map allows viewers to compare the number of retail outlets across different cities. Cities with more outlets are represented by larger circles, while cities with fewer outlets have smaller circles. This makes it easy to identify cities with a high concentration of outlets and locations where the company has a smaller presence.

Conclusion: A proportional symbol map is suitable because it combines geographic information with quantitative data, making the distribution of retail outlets easy to understand.

Map-Based Visualization of Retail Outlets

Install packages if they are not already installed

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
install.packages("ggplot2")
}
if (!requireNamespace("maps", quietly = TRUE)) {
install.packages("maps")
}
library(ggplot2)
library(maps)
```

Create sample data for 8 Indian cities

```
cities <- data.frame(
city = c("Chennai", "Bengaluru", "Hyderabad", "Mumbai",
"Delhi", "Kolkata", "Pune", "Jaipur"),
longitude = c(80.2707, 77.5946, 78.4867, 72.8777,
77.1025, 88.3639, 73.8567, 75.7873),
latitude = c(13.0827, 12.9716, 17.3850, 19.0760,
28.6139, 22.5726, 18.5204, 26.9124),
outlets = c(45, 35, 30, 60, 50, 25, 40, 20)
)
```

Load the world map

```
world_map <- map_data("world")
```

Display the map

```
ggplot() +
geom_polygon(
data = world_map,
aes(x = long, y = lat, group = group),
fill = "lightgray",
color = "white"
) +
geom_point(
data = cities,
aes(x = longitude, y = latitude, size = outlets),
color = "red",
alpha = 0.7
) +
geom_text(
data = cities,
aes(x = longitude, y = latitude, label = city),
size = 3,
nudge_y = 1
) +
coord_quickmap(
xlim = c(68, 90),
ylim = c(6, 36)
) +
scale_size_continuous(name = "Retail Outlets") +
labs(
title = "Geographic Distribution of Retail Outlets",
subtitle = "Retail outlets across 8 Indian cities",
x = "Longitude",
y = "Latitude"
) +
theme_minimal()
```



3. Network Visualizations — Network Graph

Question: Using data on which of eight websites link to each other, construct a network visualization to represent the web-link structure.

Answer:

A **Network Graph** is the most appropriate visualization for representing the connections between eight websites.

- **Nodes:** Represent the eight websites.
- **Edges:** Represent hyperlinks between websites.
- **Directed arrows:** Indicate the direction of a hyperlink from one website to another.

Explanation:

A network graph illustrates how websites are interconnected. Websites with many incoming links may be important destinations, while websites with many outgoing links may connect users to several other resources. The graph also helps identify highly connected websites, isolated websites, and groups of closely connected websites.

For example, if Website A links to Website B, an arrow is drawn from A to B.

Conclusion: A network graph is suitable because it clearly represents the relationships and link structure among multiple websites.

Network Graph of Website Links

Install required packages if not already installed

```
if (!requireNamespace("igraph", quietly = TRUE)) {
install.packages("igraph")
}
if (!requireNamespace("ggraph", quietly = TRUE)) {
install.packages("ggraph")
}
```

```
if (!requireNamespace("ggplot2", quietly = TRUE)) {
install.packages("ggplot2")
}
library(igraph)
library(ggraph)
library(ggplot2)
```

Create a list of connections between 8 websites

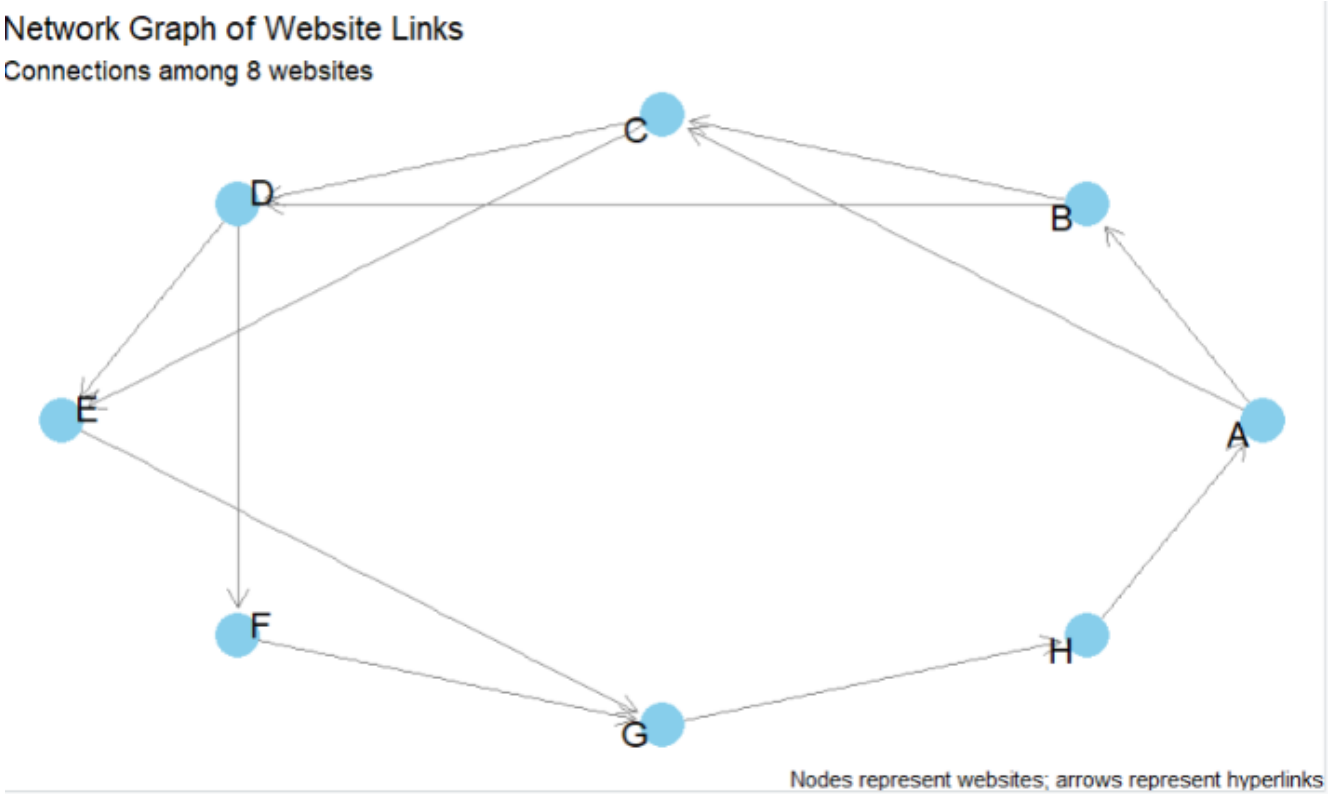
```
links <- data.frame(
from = c("A", "A", "B", "B", "C", "C", "D",
"D", "E", "F", "G", "H"),
to = c("B", "C", "C", "D", "D", "E", "E",
"F", "G", "G", "H", "A")
)
```

Create the network graph

```
website_graph <- graph_from_data_frame(
links,
directed = TRUE,
vertices = data.frame(
name = LETTERS[1:8]
)
)
```

Plot the network graph

```
ggraph(website_graph, layout = "circle") +
geom_edge_link(
arrow = arrow(length = unit(3, "mm")),
end_cap = circle(4, "mm"),
color = "gray50"
) +
geom_node_point(
color = "skyblue",
size = 8
) +
geom_node_text(
aes(label = name),
size = 5,
repel = TRUE
) +
labs(
title = "Network Graph of Website Links",
subtitle = "Connections among 8 websites",
caption = "Nodes represent websites; arrows represent hyperlinks"
) +
theme_void()
```



4. Temporal Data Representations — Line Chart

Question: Using data on the number of active users of a mobile app recorded daily over 30 days, construct an appropriate chart to show the trend over time.

Answer:

A **Line Chart** is the most appropriate visualization for showing daily active users over 30 days.

- **X-axis:** Days (Day 1 to Day 30).
- **Y-axis:** Number of active users.
- **Line:** Connects the daily user counts in chronological order.

Explanation:

A line chart makes it easy to observe changes in the number of active users over time. An upward trend indicates increasing app usage, while a downward trend suggests declining activity. Peaks may indicate days with unusually high engagement, whereas sudden drops may suggest reduced activity or other events affecting usage.

Conclusion: A line chart is suitable because it effectively represents time-series data and reveals trends, fluctuations, and changes in user activity over the 30-day period.

Question 4: Temporal Data Representations

Line Chart of Daily Active Users

Create data for 30 days

```
day <- 1:30

active_users <- c(
120, 135, 128, 145, 150, 160, 155, 170, 180, 175,
190, 200, 195, 210, 220, 215, 230, 240, 235, 250,
260, 255, 270, 280, 275, 290, 300, 295, 310, 325
)
```

Combine the data into a data frame

```
app_data <- data.frame(day, active_users)
```

Display the data

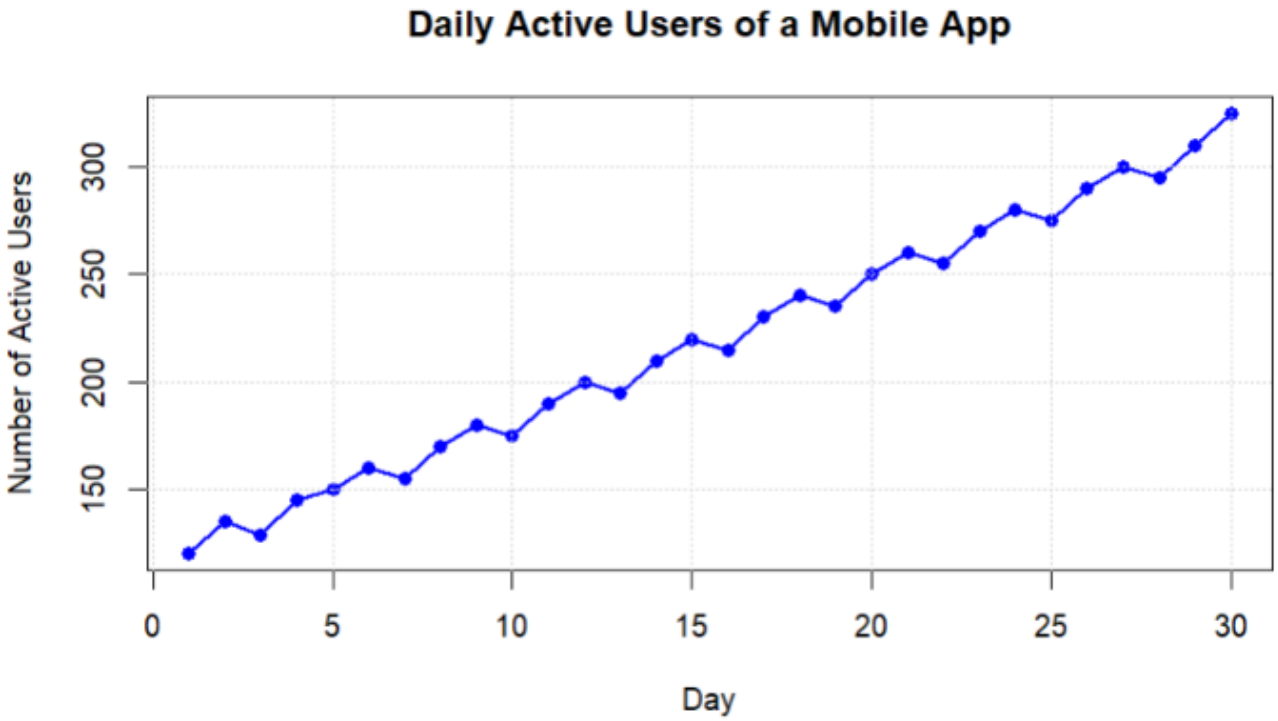
```
print(app_data)
```

Create a line chart

```
plot(
app_data$day,
app_data$active_users,
type = "o",
col = "blue",
pch = 16,
lwd = 2,
main = "Daily Active Users of a Mobile App",
xlab = "Day",
ylab = "Number of Active Users"
)
```

Add a grid

```
grid()
```



5. Multivariate Data Visualization — Parallel Coordinates Plot

Question: Using data on six students with marks in Mathematics, Science, English, and Attendance (%), construct a multivariate chart to compare them.

Answer:

A **Parallel Coordinates Plot** is an appropriate visualization for comparing the performance of six students across four variables.

- **Vertical axes:** Represent Mathematics, Science, English, and Attendance (%).
- **Lines:** Each line represents one student.
- **Values:** Each line crosses the four axes at the corresponding value for that student.

Explanation:

A parallel coordinates plot enables comparisons across several variables simultaneously. It helps identify students who perform consistently well across subjects, students who excel in a particular

subject, and students whose attendance differs from their academic performance.

For example, a student whose line passes through high values on all four axes demonstrates strong academic performance and high attendance.

Conclusion: A parallel coordinates plot is suitable because it displays multiple numerical variables simultaneously, allowing meaningful comparisons among the six students.

Parallel Coordinates Plot for 6 Students

Create sample data for 6 students

```
students <- data.frame(
Student = c("Student A", "Student B", "Student C",
"Student D", "Student E", "Student F"),
Mathematics = c(85, 70, 92, 65, 78, 88),
Science = c(80, 75, 95, 60, 82, 85),
English = c(78, 82, 88, 70, 75, 90),
Attendance = c(95, 85, 98, 75, 88, 92)
)
```

Display the data

```
print(students)
```

Convert the numerical columns to a matrix

```
marks <- as.matrix(students[, 2:5])
```

Normalize values to a scale from 0 to 100

```
normalized_marks <- apply(marks, 2, function(x) {
(x - min(x)) / (max(x) - min(x)) * 100
})
```

Create the parallel coordinates plot

```
matplot(
x = 1:4,
y = t(normalized_marks),
type = "l",
lty = 1,
lwd = 2,
col = 1:6,
xaxt = "n",
xlab = "Subjects and Attendance",
ylab = "Normalized Score (0-100)",
main = "Parallel Coordinates Plot of Student Performance"
)
```

Add labels to the horizontal axis

```
axis(
side = 1,
at = 1:4,
labels = c("Mathematics", "Science", "English", "Attendance")
)
```

Add a legend

```
legend(
"topleft",
legend = students$Student,
col = 1:6,
```

lty = 1,
lwd = 2,
cex = 0.7
)

Add grid lines

grid()

